

(PP-Short Questions)

What is a heuristic?

A heuristic algorithm is one that is used to solve a problem in a faster manner. It is more efficient than classic methods or to find an approximate solution when classic methods cannot by sacrificing optimality, accuracy or speed or completeness. It is often used to solve NP-complete problems.

Dynamic - Programming

Divide & Conquer VS Greedy approach

Follow - bottom-up approach. It generates many decision sequence as optimal solution. It always generate optimal solution. more power-technique can generate as compared to time complexity eg - 0/1 knapsack, Travelling sales man, etc.	- Follow top-down approach. - It generate only one as optimal solution. - It may or may not generate optimal sol. - Less powerful
	- Knapsack - fractional. - Activity selection problem - Minimal color etc.

(PP-SQ)

Why Dijkstra's does not work with negative weight cycle?

⇒ Because in each iteration, the algorithm only updates the answer for the nodes in the queue.

⇒ So, Dijkstra's algorithm does not recompute a node once it marks it as visited even if a shorter path exists than the previous one.

⇒ Hence, Dijkstra's algorithm fails in graphs with negative edge weights.

What are conditions when we use dynamic programming instead of greedy approach.

⇒ Greedy algorithms are often used in situations where there is a limited amount of information available.

⇒ For example, a greedy algorithm might be used to solve a problem in which the number of possible solutions is limited.

⇒ Dynamic programming is typically used when there is more information available.

(PP-5Q)

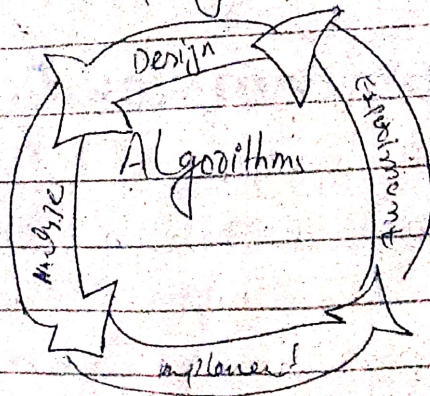
Efficiency Comparison

Running time	function class	Fast and high efficiency
1	constant	↓
$\log n$	logarithmic	
n	Linear	
n^2	quadratic	
2^n	exponential	

Q# what do you mean by daa?

Design and analysis of algorithm help to design the algorithm for solving different types of problems in Computer science.

it also helps to design and analyze the logic on how the programs will work before developing the actual code for a program.



Past-Paper (Short Question)

What is algorithm Optimality?

An algorithm can be said to be optimal if the function that describes its time complexity in the worst case is a lower bound of the function that describes the time complexity in the worst case of a problem.

What are the different complexities according to which we analyze an algorithm?

The complexity of an algorithm is divided into two parts:

* Time Complexity * Space Complexity

Write down three cases of Master method?

Already solved!

Write an algorithm using recursive function to find sum of n numbers.

Sum of Integer (n)

If $n = 1$
return 1

return (sum + sumofintegers(n-1))

What are the drawbacks of dynamic algorithm?

- 1) Can be applied to only those problems that observe the principle of optimality.
- 2) When using dynamic programming, the number of partial solutions must be saved for the direct use of results that cause increase in storage.
- 3) No general formation of dynamic program is available. Each problem has to be solved its own way.
- 4) Obviously, the space complexity increases while using dynamic programming. This is because dividing problem in sub-problem and storing intermediate result consumes memory.

What are the advantages of dynamic programming?

- 1) With DP, both local and optimal solution can be obtained.
- 2) Reduces the time complexity of problem.
- 3) Suitable for linear or non-linear problems, discrete or continuous variables and

problem. As knowledge can be gained the efficiency of dynamic programming.

Why do we use algorithms

Algorithms are used to find the best way to solve a problem based on

- Data storage
- Sorting and processing
- Machine learning

In doing so, they improve the efficiency of a program.

Algorithms are used in all areas of computing.

What properties does a problem hold for applying dynamic programming approach?

Dynamic programming can be applied to those problems which satisfy the two characteristics:

- 1) **Overlapping subproblems** - means a big problem is divided into the sub-problem, one or more subproblems may occur multiple time. ^{while}

A) Optimal substructure property - Any problem is said to be having optimal substructure property if the optimal solution of the problem can be obtained from the optimal solutions of the sub-problems.

Define LCS problem.
Already solved.

What is worst case efficiency?
- it can be determine using big-oh, which determine the set of functions grows slower than or at the same rate as the given expression.

What is average case efficiency?
Already solved.

What is asymptotic complexity?
Already solved.

What are recurrence relations?
Already solved.

Define Big-oh notation.
As

Steps for designing DP?
As

Main idea of DP approach?
As

is divide and conquer approach?

Already solved

Describe BST.

Already solved

Describe BST.

What are the different criteria used to improve the effectiveness of algorithm?

algorithm must satisfy the following criteria.

- 0 Zero or more input values.
- 0 One or more output values.
- 0 Clear and unambiguous instructions.
- 0 finiteness

What is meant by problems of optimality?

A problem is said to satisfy the principle of optimality if the sub-solutions of an optimal solution of the problems are themselves optimal solutions for their subproblems.

e.g The shortest path satisfy the principle of optimality.

it is the basic principle of DP.

Write down ingredients of dynamic programming?
Already solved.

Write the recurrence relation of quicksort.
Already solved.

Is Space Complexity affected by size of cache memory and size of registers?

No, it is affected by no. of instructions and data to execute a program.

What is good solution?

Good solution have the following characteristics:

- Resolve the contradiction
- Increase the identity of the system
- Use idle, easily available resources.

Give example where recursion is used?

Classical example for calculating factorial.

What is recursion?

What is sorting?

What is time complexity?

What is divide & conquer approach?

What is big-oh notations?

What are the red-black trees?

What is a heap?

A heap is a tree-based data structure in which all the nodes of the tree are in a specific order.

In heap the tree is complete binary tree.

Used primarily for sorting and implementing priority queues and in most famous algorithm that is dijkstra's algorithm.

There are two types of heap:

- Max-heap
- Min-heap

Heap data structure has two properties:

- Ordering
- Structural

What is fast algorithm?

Define or write Steps of designing of an algorithm?

- Design - identify the problem and understand it.
- Analyze
- implement
- Experiment

Define role of algorithm in Computing?

- To improve the efficiency of computer program
- Proper utilization of resources.

Why we use substitution method for Solving Recurrences?

This method is useful when we encounter recurrences that are non-trivial and unreadable via the master theorem.

• Then this method is useful to establish both upper and lower bounds on recurrences.

• Steps:

- Guess the form of the solution
- Use induction to show that the guess is valid.

Difference between bucket sort and Radix Sort

Bucket Sort

Bucket sort goes from MSD to

Lesser than radix sort

Efficient for 'Dense' array

$O(n+k)$ Time complexity

Used when input array

values between $[0, 1]$

Radix Sort

Sort digit by digit

Radix sort can go in both directions (LSB or MSD)

Less faster than bucket

efficient for 'Sparse' array

$O(d(n+k))$

used for integer input when keys in a certain range

What is performance analysis?

Process of calculating space and time required by that algorithm

Performance analysis of an algorithm depends on two factors:

- o Space complexity
- o Time Complexity
- o Simplicity & Creativity

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Difference between tabulation and memorization.

In dynamic programming, solving the results of sub-problems is implemented by making use of one of the two techniques

- Tabulation

- Not better than memorization

- Bottom up approach

- Solve all related sub-problem even if are not used in original solution so,

- Memory occupation is more

- faster than memorization

- Memorization

- Better than tabulation

- top-down approach

- Store only those results which have been used.

- Stack overflow may be there if there are too many recursive call.

What is tabulation and memorization?

Tabulation:

Tabulation is the bottom-up approach which starts with solving

lowest level of sub-problems. The solution of the solved sub-problem is stored in the tabular form.

The solution of these sub-problems move up to the next level of sub-problems and so on upto the solution of the whole problem.

Memorization:

Memorization is top-down approach which starts from the highest level of sub-problems which are closed to the original problem.

This method recursively calls the next level of subproblems and so on

Why we analyze algorithm?

- Evaluate algorithm performance
- Compare different algorithm

Analyze what about Mem,

- Running time, memory usage, solution quality,
- worst case and "typical" case

#

Dynamic Programming

V/S

Divide-&-Conquer

1) Used to solve optimization problem

2) follow bottom-up approach

3) Sub-problems are dependent

4) The solution of sub-problem compute once and store in table for later use.

Examples:

o 0/1 knapsack, all pairs shortest path, LCS, matrix-chain multiplication etc

1) used to solve decision problem

2) follow top-down approach

3) Sub-problems are independent

5) Solution of sub-problem compute recursively more than once.

o Quick-sort

o Merge-sort

o Strassen matrix

Similarities: DP and DAC

o Both technique divide the problem into subproblem.

o Both are recursive in nature.

Similarities B/w DP and GT:-

o Both are used to generate optimal solution.

o Decision at each stage.

Complexity classes:

As, there exist some problems whose solutions are not yet found. Problems are divided into classes known as complexity classes.

Also called research oriented problem.

P-Problems:

The P class stands for polynomial Time.

It is the collection of decision problem that can be solved by deterministic machine in polynomial time.

NP-Problems:

- The set of all decision-based problems came into the division of NP problems who cannot be solved or produced an output within polynomial time but verified in the polynomial time.

- NP class contains P class as a subset.

- NP problems are being hard to solve.

- Also called non-deterministic.

Determin

2

Deterministic:

- An algorithm is called deterministic algorithm if the compiler for it produce the same output going through the same states for a particular input

e.g. BS, LS

nonDeterministic:

- For the same input, the compiler may produce different output in different runs.

e.g. 0/1 knapsack

Knapsack Problem:-

You are given the following:-

- A knapsack with limited weight capacity.
- Few items each having some weight and value.
- The problem states:-
 - which item should be placed into the knapsack such that:-
 - o The value or profit obtained by putting the items into the KS is maximum.
 - And ^{the} weight limit of the knapsack does not exceed.

Knapsack Problem Variants

- o Fractional knapsack problem
- o 0/1 knapsack problem

Fractional Knapsack

- it is a technique that is used to solve knapsack problem.
- In fractional knapsack, the items are broken in order to maximize the profit.
- The problem in which we break the problem or item is known as fractional knapsack problem.
- These problems can be solved using:
 - o Brute-force
 - o Greedy algorithm.